DE–2742

Sub. Code
11

DISTANCE EDUCATION

M.C.A. DEGREE EXAMINATION, DECEMBER 2017.

First Semester

PRINCIPLES OF INFORMATION TECHNOLOGY

(2005 to 2010 Calendar Year)

Time : Three hours Maximum : 100 marks

Answer any FIVE questions.

All questions carry equal marks.

(5 × 20 = 100)

1. (a) Describe the importance of "software" element and "Hardware" elements in computer technology era. (14)

   (b) Discuss on ethics of information technology. (6)

2. Write notes on the following with necessary examples.

   (a) Word processing. (12)

   (b) Presentation graphics software. (8)

3. Explain the database software that are very useful to everyone. (20)

4. Give a brief account on:

   (a) Web Browsers. (10)

   (b) Desktop accessories and personal information managers. (10)

5. (a) Compare and contrast internet with intranet. (12)

   (b) Give a note on Integrated services digital network. (8)

6. Narrate the concept of communication networks with its topology. (20)

7. Write notes on:

   (a) Compression and decompression. (8)

   (b) Different kinds of secondary storage devices. (12)

8. (a) List and explain the six phases of system analysis and design. (10)

   (b) Explain the concept of object oriented and visual programming languages. (10)

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DE–2743

Sub. Code
12

DISTANCE EDUCATION

M.C.A. DEGREE EXAMINATION, DECEMBER 2017.

First Semester

DIGITAL COMPUTER ORGANIZATION

(2005 to 2010 Calendar Year)

Time : Three hours Maximum : 100 marks

Answer any FIVE questions.

(5 × 20 = 100)

1. (a) Convert the decimal number 153.513 into binary and octal.

   (b) Perform the subtraction using 2's complement and 1's complement 753 – 864.

2. (a) Simplify :

   (i) $AB + \overline{A}C + A\overline{B}C\ (A \cdot B + C)$

   (ii) $\overline{A}\,\overline{B}\,CD + ABCD + \overline{A}C\overline{D} + \overline{A}\,CD + A\overline{B}D$.

   (b) Using Karnaugh map, reduce the following expression in sum of products and products of sums forms :

   $F = \Sigma(1, 5, 6, 7, 14, 15)$.

3. (a) Design a Half adder and explain.

   (b) Explain how can you perform addition and subtraction in a parallel arithmetic element.

4. (a) Explain the working principle of clocked RS flip-flop.

   (b) Write about shift registers.

5. (a) Discuss about fetch and execute cycle.

   (b) Describe the sequence of operations of control registers.

6. (a) What are interrupts? How priority is established?

   (b) Describe the organization of IOP.

7. (a) Design a BCD counter and explain.

   (b) Explain the need of secondary memory.

8. (a) Draw the block diagram of decoder and explain.

   (b) Write about character codes.

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DE–2744

Sub. Code
13

DISTANCE EDUCATION

M.C.A. DEGREE EXAMINATION, DECEMBER 2017.

First Semester

C AND DATA STRUCTURES

(2005 to 2010 CalendarYear)

Time : Three hours Maximum : 100 marks

Answer any FIVE questions.

All questions carry equal marks.

($5 \times 20 = 100$)

1. (a) Give the character set of C. (5)

   (b) Explain the execution of switch statement with examples. (15)

---------- Cut here ----------

2. (a) Define a Function. How to access it? (10)

   (b) How to pass an array to functions? (10)

---------- Cut here ----------

3. (a) Define Pointers. Give the importance of it. (10)

   (b) Write a program in C to create a data file. (10)

4. (a) What is the need for structuring data? (10)

(b) Explain the following operations in a stack

(i) POP (ii) PUSH. (10)

---------- Cut here ----------

5. (a) How to search an element in a linked list? (10)

(b) Discuss on any one application of a tree. (10)

---------- Cut here ----------

6. (a) Explain the following functions :

(i) gets

(ii) puts. (10)

(b) Write a program in C to find the average of N numbers. (where N is a non-negative integer and given as an input) (10)

---------- Cut here ----------

7. (a) Discuss the concept of C preprocessor. (10)

(b) Write a program in C to check whether the given string in Palindrome or not. (10)

---------- Cut here ----------

8. (a) How to delete an item in a Queue? (10)

(b) What is meant by a binary tree? How to represent a binary tree? (10)

DE–2745

Sub. Code
14

DISTANCE EDUCATION

M.C.A. DEGREE EXAMINATION, DECEMBER 2017.

First Semester

OFFICE AUTOMATION

(2005 to 2010 Calendar Year)

Time : Three hours Maximum : 100 marks

Answer any FIVE questions.

All questions carry equal marks.

($5 \times 20 = 100$)

1. (a) Explain how to manage files with windows explorer.

   (b) Explain creating opening and saving files in MS-Office.

2. (a) Explain the editing keys in MS-Word.

   (b) Explain formatting in MS-Word with an example.

3. Explain the following in MS-Word.

   (a) Spell checking

   (b) Align text

   (c) Indent text

   (d) Create bullet and numbed list.

4. Explain the following in MS-EXCEL.

   (a) entering formulas

   (b) absolute addresses in formulas

   (c) working with ranges

   (d) move and copy data

5. (a) Explain the various aspects of building an EXCEL database.

   (b) Explain the use of charts in MS-EXCEL with examples.

6. (a) Explain how to change views in power point.

   (b) Explain the procedure for formatting and running a presentation.

7. Explain the following in powerpoint

   (a) drawing on slides

   (b) color scheme

   (c) design template

   (d) view slide show.

8. (a) Explain the applications of MS-ACCESS.

   (b) Explain how to create, save and edit a query in MS-ACCESS.

DE–9830

Sub. Code
15

DISTANCE EDUCATION

M.C.A. DEGREE EXAMINATION, DECEMBER 2017.

First Semester

Computer Applications

LAB I — DATA STRUCTURES USING 'C'

(2005 to 2010 Calendar Year)

Time : Three hours Maximum : 100 marks

Break-Up of Marks

Record Note book	:	05
Algorithm/Flowcharts	:	15 (7 + 8)
Program (s)	:	35 (17 + 18)
Debugging/Execution	:	35 (17 + 18)
Results/Outputs	:	10 (5 + 5)
Total	:	100

Candidate has to answer ONE question chosen by lot system.

1. (a) Write a C program to find that the given word is a palindrome or not. (LIRIL is a palindrome, HAMAM is not a palindrome).

   (b) Write a C program to sort a given set of '*n*' integers in ascending order. (40 + 60)

2. (a) Write a C program to find the value of nCr using function

$(nCr = \lfloor n / \lfloor r \lfloor n-r)$.

(b) Write a C program to implement 'push' and 'pop' operations on stack. (50 + 50)

---------- Cut here ----------

3. (a) Write a C program to find the roots of a quadratic equation of the form $aX^2 + bX + C = 0$.

(b) Write a C program to implement 'insert' and 'delete' operations on 'queue'. (50 + 50)

---------- Cut here ----------

4. (a) Write a complete C program to manipulate matrices such as Addition, Subtraction and Multiplication of matrices.

(b) Write a C program to copy the contents of one file to another file. (60 + 40)

---------- Cut here ----------

5. (a) Write a C program to demonstrate string operations/string manipulations.

(b) Write a C program to evaluate a given mathematical expressing using 'stack'. (50 + 50)

---------- Cut here ----------

6. (a) Write a C program to multiply two matrices and provide the compatiability check for multiplication matrices.

(b) Write a C program to implement 'stack' operations. (50 + 50)

7. (a) Write a C program using recursive function to calculate factorial of a given integer.

   (b) Write a C program to implement 'selection sort' to sort '*n*' integers in ascending order.

---------- Cut here ----------

8. (a) Write a C program to do the following :

   Enter 10 students register no, three subject marks. Declare 'PASS' or 'FAIL' depends on if all the subjects marks are greater than 39 the 'PASS' else 'FAIL'.

   (b) Write a C program to solve quadratic equation.

(60 + 40)

DE–9831

Sub. Code
16

DISTANCE EDUCATION

M.C.A. DEGREE EXAMINATION, DECEMBER 2017.

First Semester

Lab II – MS OFFICE AND INTERNET

(2005–2010 Calendar Year)

Time : Three hours Maximum : 100 marks

Breakup Marks
Record Note Book : 5
Algorithm/Flow chart : 15
Program : 35
Debugging and Execution : 35
Result : 10
Total : 100

Answer any TWO questions choosing ONE from Section A and another from Section B.

Each question carries 50 marks.

SECTION A

1. Draw Line, Bar, Pie charts for the data given below :

ABC Company Ltd.

Year	Sales	Cost	Profit
1991	1000	400	600

Add data for 10 years from 1991 to 2000. Provide titles, legends, grids and data labels.

2. Create STUDENT table with the following fields REG NO, NAME, MARK 1, MARK 2, MARK 3. Create Address table with fields REG NO, STREET, CITY and PIN.

   (a) Write a query to display REG NO, NAME and TOTAL of all the 3 subject marks.

   (b) Write a query to display REG NO, NAME, STREET, CITY, PIN and total of all the 3 subject marks.

---------- Cut here ----------

3. Create Inventory table with fields ITEM NO, NAME, QTY ON HAND, REORDER LEVEL :

   (a) Create a form in custom format.

   (b) Create a query REORDER to show all the items.

   (c) Create a macro that executes REORDER query automatically.

---------- Cut here ----------

4. Prepare your resume with your photograph inserted. Use table, bullets and different color features.

---------- Cut here ----------

5. Prepare a news report using two columns, insert a picture in the first column and make the text flow around it.

---------- Cut here ----------

6. Prepare an invitation for a function to be conducted in your institution. Use different text orientation and pictures to make it attractive.

7. Create a new document and type the following mathematical expression.

$$\int_{0}^{\infty} x \; dy/dx \sum_{i=1}^{100} ax_i + b + |A| \pm \sqrt[3]{ax + b\lambda} \; .$$

---------- Cut here ----------

8. Create a work sheet in EXCEL with the following columns.

Employer no, Name, Designation, Basicpay, HRA, DA, LIC, PF, Grosspay, Net pay

(a) Type data for Emp no, Name, Designation, Basicpay, LIC and PF.

(b) Calculate HRA = 20% of Basic

DA = 30% of Basic

Grosspay = Basic + HRA + DA

Netpay = Grosspay – (LIC + PF)

(c) Draw the bar chart between Emp name and Netpay.

(d) Sort the designation column and employee name column at a time.

---------- Cut here ----------

SECTION B

9. Create HTML page that plays video and audio files.

10. Create HTML page to display the pictures of Elephant and Monkey. Link to another page when user clicks over the picture. The linked page should display features of the animal clicked.

---------- Cut here ----------

11. Create HTML page that divides the screen space into 3 frames (one vertical and two horizontal of equal size). Display 3 different files in then with scrolling titles relevant to their contents.

DE–2746

Sub. Code
21

DISTANCE EDUCATION

M.C.A. DEGREE EXAMINATION, DECEMBER 2017.

Second Semester

OBJECT ORIENTED PROGRAMMING AND C++

(2005 to 2010 calendar year)

Time : Three hours Maximum : 100 marks

Answer any FIVE questions.

All questions carry equal marks.

($5 \times 20 = 100$)

1. (a) Write short notes on : Object Oriented Languages. (10)

   (b) Write a C++ program to find the sum and average of N input numbers. (10)

2. (a) Explain various control structures in C++ with suitable examples. (10)

   (b) Discuss briefly about various expressions in C++ with examples. (10)

3. (a) Distinguish between one dimensional and multidimensional arrays. (10)

   (b) Write short Notes on : pointers. (10)

4. (a) Distinguish between Pass-by-Value and Pass-by-Reference. (10)

   (b) What is meant by Recursion? Explain this with an example. (10)

5. (a) Write a C++ program to illustrate function overloading. (10)

   (b) What are Inline functions? Discuss them with examples. (10)

6. (a) What are constructors? List the characteristics of constructors. (10)

   (b) Discuss briefly on : Type conversions with examples. (10)

7. (a) Write a C++ program to illustrate single inheritance. (10)

   (b) List the rules for virtual functions. (10)

8. (a) Discuss briefly about the classes for file stream operations. (10)

   (b) What are command line arguments? Explain with an example C++ program. (10)

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DE–2747

Sub. Code
22

DISTANCE EDUCATION

M.C.A. DEGREE EXAMINATION, DECEMBER 2017.

Second Semester

SYSTEMS ANALYSIS AND DESIGN

(2005–2010 Calendar Year)

Time : Three hours Maximum : 100 marks

Answer any FIVE questions.

All questions carry equal marks.

(5 × 20 = 100)

1. (a) Explain the characteristics of a system.

   (b) Describe the various elements in a system.

2. Explain the various stages in the life cycle approach to system development.

3. Explain the various tools available for gathering information.

4. (a) Explain the various types of feasibility considerations.

   (b) Explain the types of costs with examples.

5. (a) Define a form.

   (b) Explain the classification of forms.

   (c) Explain the requirements of form design.

6. (a) Explain the various design methodologies with examples.

   (b) Bring out the differences between a file and a database.

7. (a) Explain with examples, the various types of flowcharts.

   (b) Explain the use of system testing with examples.

8. (a) Explain the various conversion methods.

   (b) Explain the major phases in hardware and software selection.

DE–2748

Sub. Code
23

DISTANCE EDUCATION

M.C.A. DEGREE EXAMINATION, DECEMBER 2017.

Second Semester

VISUAL PROGRAMMING – I

(2005 to 2010 calendar year)

Time : Three hours Maximum : 100 marks

Answer any FIVE questions.

All questions carry equal marks.

(5 × 20 = 100)

1. (a) What is meant by event driven programming?

   (b) Explain the VB IDE with the components and their purpose.

2. (a) Explain any five components in the Toolbox.

   (b) Write code and design a form to receive two numbers and to display the sum and calculated value in two separate text boxes.

3. (a) Explain the looping statements in VB in the flow chart and examples.

   (b) Explain the timer control with suitable example.

4. (a) Discuss any seven string functions with examples.

   (b) What is a control away? Explain.

5. (a) What is a function? What is a procedure? Explain.

   (b) Write VB code and design form to receive a number and to display whether that number is divisible by 5 and 10 by using a function.

6. (a) Explain

   (i) List and combo boxes

   (ii) Grid control.

   (b) Explain any five date function in VB.

7. (a) Explain the object oriented features in VB.

   (b) What is a model window? Explain.

   (c) Explain menu creation with examples.

8. (a) Discuss the database interaction features of VB with samples.

   (b) Write VB code and create a form to read the contents of a file and to display it.

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DE–2749

Sub. Code
24

DISTANCE EDUCATION

M.C.A. DEGREE EXAMINATION, DECEMBER 2017.

Second Semester

COMPUTER GRAPHICS

(2005 to 2010 Calendar Year)

Time : Three hours Maximum : 100 marks

Answer any FIVE questions.

All questions carry equal marks.

(5 × 20 = 100)

1. Explain the Bresenham's line drawing technique with program segment. (20)

2. (a) Compare and contrast raster scan displays with random scan displays. (10)

   (b) Describe the working principles of various hard copy devices. (10)

3. (a) Derive the matrix for two dimensional rotation transformation. (10)

   (b) With matrix representation, discuss on composite transformations. (10)

4. Give a neat description on Cohen-Sutherland line clipping technique. Also narrate the program segment. (20)

5. Write notes on the following three dimensional transformation principles with matrix representation:

   (a) Translation (5)

   (b) Scaling (7)

   (c) Mirror reflection. (8)

6. Explain the following with neat diagrams, related to three-dimentional object:

   (a) Parallel projection (10)

   (b) Perspective projection (10)

7. Discuss the following algorithms:

   (a) Depth-sorting method. (10)

   (b) Hiden surface line removal. (10)

8. Write notes on the following related to user interface design.

   (a) The command language (7)

   (b) Information display (7)

   (c) Feedback. (6)

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DE–9832

Sub. Code
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DISTANCE EDUCATION

M.C.A. DEGREE EXAMINATION, DECEMBER 2017.

Second Semester

Computer Applications

COMPUTER LAB III (GRAPHICS USING C++)

(2005 to 2010 Calendar Year Onwards)

Time : Three hours Maximum : 100 marks

Examiner has to select ONE questions and give it to the candidate by LOT system

1. (a) Write a C++ program to draw non-linear objects arcs and curve.

   (b) Write a C++ program to implement the Bresenham's algorithm for drawing circle and ellipse.

---------- Cut here ----------

2. (a) Write a C++ program to show graphical transformations of a two dimensional object.

   (b) Write a C++ program to implement the polygon clipping algorithm.

3. (a) Write a C++ program to show graphical transformations of a three dimensional object.

   (b) Write a C++ program to manipulate the graphical object using SHADING technique.

---------- Cut here ----------

4. (a) Write a C++ program to draw any 3-D object and perform animation on it.

   (b) Write a C++ program to draw a straight line using DDA technique.

---------- Cut here ----------

5. (a) Write a C++ program to draw non-linear object ellipse.

   (b) Write C++ program to draw a circle. Divide it into four quadratic and fillup different color for each quadrant.

---------- Cut here ----------

6. (a) Write a C++ program to draw non-linear object circle and three concentric circle.

   (b) Write a C++ program to draw a ball and perform animation on it.

DE–9833

Sub. Code
26

DISTANCE EDUCATION

M.C.A. DEGREE EXAMINATION, DECEMBER 2017.

Second Semester

Computer Lab IV – VISUAL BASIC

(2005 to 2010 Calendar year)

Time : Three hours Maximum : 100 marks

Examiner should select and give ONE question to each candidate by LOT system. Each subdivision carries 50 marks.

Break up :
Record : 5
Algorithm/Flow chart : 15
Program : 35
Debugging and Execution : 35
Result : 10
Total : 100

1. (a) Write a VB Project to receive three numbers in text boxes and to display the largest and the smallest numbers.

   (b) Write a VB Project to display the records from a student table with fields number, name, subject and mark. Also display the result as 'pass' for each subject if the mark is greater than 49.

2. (a) Write a VB Project to display the items in two different lists and to move the elements one-by-one between the lists based on timer.

(b) Write VB code and design a form to display a question in a text box and four different choices with radio buttons. if the correct answer is checked, display a message suitably.

---------- Cut here ----------

3. (a) Write a VB Project to create an active X control to design a text box that receives only alphabets. If any other key is pressed, suitable error manage to be displayed.

(b) Write a VB Project to read *n* numbers and to display the sorted numbers.

---------- Cut here ----------

4. (a) Write a VB Project to design a calculated that implements addition, subtraction, multiplication and division.

(b) Write a VB Project to receive a number and to display its multiplication table.

---------- Cut here ----------

5. (a) Write a VB Project to read the employee details and to insert them into a data base. Also add the features for updation and deletion.

(b) Write a VB Project to create a screen saver that displays alphabets one by one.

6. (a) Write a VB Project to read a string and to display whether it is a Palindrome or not.

(b) Write a VB Project to design an editor that performs the basic functions of an editor.

---------- Cut here ----------

7. (a) Write a VB Project to read and display 10 records, from a data base, at a time, provide facilities for scrolling.

(b) Write a VB Project to display four images at a time out of 10 images. The images are to revolve, that is, if one image is moved out, another are comes in. This is to be performed with a 1 second delay.

---------- Cut here ----------

8. (a) Write a VB Project to read a year and to display whether it is a leap year or not.

(b) Write a VB Project to read the key value and to retrieve the corresponding record from the data base.

DE–2750

Sub. Code
31

DISTANCE EDUCATION

M.C.A. (S)/M.C.A (Lateral) DEGREE EXAMINATION, DECEMBER 2017.

Third Semester

DISCRETE MATHEMATICS

(2005 to 2010 Calendar Year)

Time : Three hours Maximum : 100 marks

Answer any FIVE questions.

$(5 \times 20 = 100)$

1. (a) Show that $\left(\neg P \wedge \left(\neg Q \wedge R\right)\right) \cup (Q \cap R) \vee (P \cap R) \Leftrightarrow R$.

   (b) Find the principal disjunctive normal form of the formula

   $\left(\neg((P \vee Q) \wedge R)\right) \wedge (P \vee R)$.

2. (a) Show that $(\forall x)(P(x) \wedge Q(x)) \leftrightarrow ((\forall x)P(x)) \wedge (\forall x)(Q(x))$.

   (b) Show that $Q \vee \left(P \wedge \neg Q\right) \vee \left(\neg P \wedge \neg Q\right)$ is a tautology.

3. (a) For any three sets A, B, C; prove that $A \cup (B \cap C) = (A \cup B) \cap (A \cup C)$.

   (b) Prove that the relation "Congruence modulo m" over the set of positive integer is an equivalence relation.

4. (a) If $f : A \to B$ and $g : B \to C$ are bijective mappings, prove that $g.f : A \to C$ is also bijective.

   (b) Let $f : A \to B$ and $g : B \to C$ be both one-to-one and onto functions. Prove that $(g.f)^{-1} = f^{-1} \cdot g^{-1}$.

5. (a) Prove that for any commutative monoid $(M,*)$, the set of idempotent elements of M forms a submonoid.

   (b) Let G be an abelian group, prove that for all $a, b \in G(a * b)^n = a^n * b^n$ for every integer n.

6. (a) Let H and K be two subgroups of G, prove that HK is a subgroup of G if and only if HK = KH.

   (b) State and prove Lagrange's theorem.

7. (a) Define : (i) Complete graph (ii) Bipartite graph (iii) Adjacency matrix (iv) Incidence matrix (v) Path matrix.

   (b) Prove that in any graph, the number of vertices of odd degree is even.

8. (a) Prove that a tree with n vertices has $(n-1)$ edges.

   (b) Prove that a simple graph with n vertices and k components can have at most $(n-k)(n-k+1)/2$ edges.

DE–2751

Sub. Code
32

DISTANCE EDUCATION

M.C.A./M.C.A. (Lateral) DEGREE EXAMINATION, DECEMBER 2017.

Third Semester

OPERATING SYSTEMS

(2005 to 2010 Calendar year)

Time : Three hours Maximum : 100 marks

Answer any FIVE questions.

All questions carry equal marks.

(5 × 20 = 100)

1. Write notes on the following operating systems:

   (a) Mainframe systems. (6)

   (b) Multiprocessor systems. (7)

   (c) Distributed systems. (7)

2. Explain the various operating system services with system structure and programs. (20)

3. (a) Give a note on critical section problem. (10)

   (b) Write about event counters and monitors. (10)

4. How does deadlock occurrence is detected and prevented? Explain with suitable illustration. (20)

5. (a) Elucidate the concept of deadlock characterization. (10)

   (b) Narrate the paging technique with its hardware implementation procedure. (10)

6. (a) Discuss the process of managing memory with linked list and buddy system. (10)

   (b) Elaborate the segmentation technique that is used for memory management. (10)

7. Write notes on the following page replacement techniques:

   (a) FIFO page replacement. (8)

   (b) LRU approximation page replacement. (12)

8. Explain the following related to file management system:

   (a) File storage. (6)

   (b) File system reliability. (6)

   (c) File system performance. (8)

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DE–2752

Sub. Code
33

DISTANCE EDUCATION

M.C.A./M.C.A. (Lateral) DEGREE EXAMINATION, DECEMBER 2017.

Third Semester

RDBMS

(2005 to 2010 Calendar Year)

Time : Three hours Maximum : 100 marks

Answer any FIVE questions.

All questions carry equal marks.

($5 \times 20 = 100$)

1. (a) Explain the concept of Database with an example.

   (b) Explain in detail about Schema and Subschema.

2. (a) Describe the features and advantages of DBMS.

   (b) Explain the mapping operations on Relational Model.

3. (a) Discuss on Data Definition facilities in DBMS.

   (b) Explain the concept of Relational model.

4. (a) Explain the Architecture of Oracle Database.

   (b) Explain in detail about Client/Server systems.

5. (a) Explain the following: i. Physical files ii. Network systems.

   (b) Explain Data Access SQL Commands in Oracle.

6. (a) How to join multiple tables in Oracle? Explain.

   (b) Explain Subqueries in Oracle with examples.

7. (a) Explain the following: (i) Create table (ii) Update table.

   (b) Explain indexes in Oracle.

8. (a) Explain the need for database objects in Oracle.

   (b) Write a detailed note on Embedded SQL statements.

DE–2753

Sub. Code
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DISTANCE EDUCATION

M.C.A./M.C.A. (Lateral) DEGREE EXAMINATION, DECEMBER 2017.

Third Semester

Computer Applications

UNIX AND SHELL PROGRAMMING

(2005-2010 Calendar Year)

Time : Three hours Maximum : 100 marks

Answer any FIVE questions.

All questions carry equal marks.

(5 × 20 = 100)

1. (a) Discuss the features of 'Sed'.

   (b) Explain the features of 'awk'. (10 + 10)

2. (a) Explain the Directory related unix commands.

   (b) Explain the 'file permissions' concept with examples. (10 +10)

3. Explain the following compound unix commands with suitable examples.

   (a) Redirection

   (b) Pipes

   (c) Filters

   (d) Tee. (5+5+5+5)

4. (a) What are shell variables and shell scripts?

   (b) Discuss the shell meta characters and its purpose.

(10+10)

5. (a) Explain the looping statements in shell programming.

   (b) Write a note on rest, exit and trap commands. (14+6)

6. (a) What are 'Procedures' in shell programming? Explain.

   (b) Write a note on document 'preparation tools'.(10+10)

7. (a) What are 'unix system cells'?

   (b) Explain the debugging facility provided in unix.

(10+10)

8. (a) Write a shell script of find the biggest of three numbers using relational operators.

   (b) Write the purpose of 'expr' and 'eval' commands.

(10+10)

DE–2754

Sub. Code
35

DISTANCE EDUCATION

M.C.A./ MCA (S) Lateral DEGREE EXAMINATION, DECEMBER 2017.

Third Semester

COMMUNICATION SKILLS

(2005 to 2010 Calendar Year)

Time : Three hours Maximum : 100 marks

Answer any FIVE questions

ALL questions carry equal marks.

($5 \times 20 = 100$)

1. (a) Explain the process of communication.

   (b) Differentiate between the interpersonal communication and intrapersonal communication.

2. (a) Give any five 'Do's and Dont's in an interview.

   (b) How can the Mock-interview help in the preparation for the real interview?

3. (a) Write on the various skills assessed during the group discussion process.

   (b) Make an analysis of the various body languages.

4. (a) Bring out importance of Negotiation.

   (b) Write an essay on how a meeting scheduled, meeting information distributed and conducted.

5. (a) Make notes on :

   (i) Upward communication and

   (ii) Downward communication.

   (b) How is a presentation designed?

6. (a) Make a review of Reading techniques.

   (b) Give the tips for Article writing.

7. (a) Write on the basic Listening Modes.

   (b) Write briefly on :

   (i) Conversation starters

   (ii) Conversation keepers and

   (iii) Conversation stoppers.

8. (a) What are important rules for a successful brainstorming?

   (b) How can speaking ability be developed?

DE–9834

Sub. Code
36

DISTANCE EDUCATION

M.C.A./M.C.A. (Lateral) DEGREE EXAMINATION, DECEMBER 2017.

Third Semester

LAB V – RDBMS

(2005 to 2010 calendar year)

Time : Three hours Maximum : 100 marks

ONE Question is to be given by the Examiner to each candidate by LOT system

Break-up marks

Record:	5
Algorithm:	15
Program :	35
Debugging and Execution :	35
Result :	10
	100

1. Write a PLSQL program to prepare a mark statement of a student using

   (a) STUDENT TABLE AND MARK TABLE Assume your own data.

   (b) Write an SQL program to do the following

   Assign a letter grade to each grade based on the average score and list

Register number, Average Score, Grade. The minimum pass for each subject is 50.

The Grading System :

Average Score	Grade
90–100	A
75-89	B
60–74	C
50–59	D
0–49	F

---------- Cut here ----------

2. (a) A hospital maintains blood donors' record information in a file. The file contains the following fields: Donor number, Donor name, Donor age, Donor address, Place of birth, Blood group.

Write a program using Oracle to print out the number, name and address of the donors for the following categories:

(i) Blood donors having A blood group

(ii) Blood donors in the age group between 18 to 25

(iii) Male donors with 'B' Blood group in the age between 18 to 25.

(b) An airline reservation database contains the reservation table and personal table.

Reservation Table contains following fields:

Flight number, class, passenger name, seat number

Personal table contains following fields:

Passenger name, sex. Age, Nationality

Write a program in ORACLE to prepare the following reports:

(i) List the passengers names with seat number according to class and flight number.

(ii) Total number of married female passengers in a particular flight.

(iii) List out the male passengers in the age between 20 to 28 for all flights

---------- Cut here ----------

3. (a) A news paper vendor buys newspapers on wholesale from a distributor for Rs. 4 and sell them in retail for Rs. 4.50. At the end of the day the unsold papers are returned to the distributor for Rs. 1 rebate per paper. Write a program using Oracle to prepare a report for the newspaper vendor in a neat format.

(b) Write a program in SQL to prepare salary report for 5 employees. Assume your own database.

4. (a) Write a program in SQL for hospital billing system with the following fields.

Patient number, name, age, date of birth, attending doctor, patient type, consulting charge, blood test charge, x-ray charge and total fee.

Prepare reports with the following conditions :

(i) The patients undergone blood test.

(ii) The patients who belong to in-patient category.

(b) Write a SQL program to prepare a bill for canteen with two tables namely MENU and BILL. Assume your own database.

DE–9835

Sub. Code
37

DISTANCE EDUCATION

M.C.A./M.C.A. (Lateral) DEGREE EXAMINATION, DECEMBER 2017.

Third Semester

Lab VI — ALGORITHM AND SHELL PROGRAMMING

(2005 to 2010 Calendar Year)

Time : Three hours Maximum : 100 marks

One question is to be given by the examiner to each candidate by Lot System.

Break-up marks

Record Note Book	: 5
Algorithm/Flow chart	: 15
Program	: 35
Debugging and Execution	: 35
Result	: 10
Total	: 100

1. (a) Write a recursive routine for printing the first 'N' Fibonacci numbers.

   (b) Write a shell program to prepare the electricity bill for the consumers of type domestic based on the following constraints :

   For the first 100 units – Nil

   From 101 to 200 units – Rs. 1.50/- per unit + Surcharge Rs. 20/-

   From 201 to 500 units – Rs. 2.25/- per unit + surcharge Rs. 30/-

   Above 500 – Rs. 3/- per unit + surcharge Rs. 50/-

   Print the electricity bill in a pleasing format.

2. (a) Write a recursive algorithm for Quick Sort.

   (b) Write a shell program to accept a string from the standard input device and print it using reverse video mode. Also, display it using bold and italic style.

---------- Cut here ----------

3. (a) Write an algorithm for implementing the Towers of Hanoi problem.

   (b) Write a shell program to simulate a calculator using yacc command.

---------- Cut here ----------

4. (a) Using the backtracking methodology, write an algorithm for Knight tour problem.

   (b) Write a shell program to find the sum of the following series :

$$\frac{1}{1!}+\frac{1}{2!}+\frac{1}{3!}+\cdots+\frac{1}{10!}.$$

---------- Cut here ----------

5. (a) Write an algorithm for arranging the numbers in ascending order using the straight selection sort.

   (b) Write a shell program to print the monthly payslip for the employees of an organization.

---------- Cut here ----------

6. (a) Write an algorithm for the travelling sales person problem using branch and bound method.

   (b) Write a menu driven shell program to perform the following tasks :

      (i) To list all the files

      (ii) To print the today's date.

7. (a) Write an algorithm for heap sorting method.

   (b) Write a menu driven shell program to copy, edit, rename, and delete a file.

---------- Cut here ----------

8. (a) Write an algorithm for finding the factorial of a given integer.

   (b) An hospital maintains the details of patients like patient id. name of the patient, their age and blood group. Write a shell program to list the details of the patients according to blood group and to list all patient details between the age limit of 40 to 50.

DE–2755

Sub. Code
41

DISTANCE EDUCATION

M.C.A./M.C.A. (Lateral) DEGREE EXAMINATION, DECEMBER 2017.

Fourth Semester

COMPUTER NETWORKS

(2005 to 2010 Calendar Year)

Time : Three hours Maximum : 100 marks

Answer any FIVE questions only.

All questions carry equal marks.

($5 \times 20 = 100$)

1. (a) What is a Network? What are the components of a data communication network? Also mention any two uses of a network. (1 + 5 + 4)

   (b) Explain the layered architecture of OSI network model. (10)

2. (a) Write an essay on guided transmission media. (10)

   (b) Narrate on the communication satellites. (10)

3. (a) Explain the four methods used by Data Link Layer to transmit a good frame. (10)

   (b) With an example, explain role of CRC in detecting errors in communication. (10)

4. (a) How does HDLC support both half and full duplex modes in point-to-point and multipoint configurations? (12)

   (b) Write a note on firewalls. (8)

5. (a) With an example, explain the Link State Routing in detail. (12)

   (b) Compare ARP with RARP. (8)

6. Give an account on the duties of transport layer in transmission. (20)

7. (a) Write an essay on Transmission Control Protocol. (10)

   (b) Write notes on :

      (i) Domain Naming System and

      (ii) SNMP. (5 + 5)

8. (a) Write an essay on Electronic Mail. (12)

   (b) Write short notes on :

      (i) URL and

      (ii) Audio compression. (4 + 4)

DE–2756

Sub. Code
42

DISTANCE EDUCATION

M.C.A./M.C.A. (Lateral) DEGREE EXAMINATION, DECEMBER 2017.

Fourth Semester

ACCOUNTING AND FINANCIAL MANAGEMENT

(2005 to 2010 Calendar Year)

Time : Three hours Maximum : 100 marks

Answer any FIVE questions.

All questions carry equal marks.

1. Explain different types of errors with appropriate examples.

2. Viswa's trial balance appeared as follows on 31st December 2013.

	Rs.
Capital	92,000
Creditors	18,852
Bills payable	6,930
Sales	1,21,850
Reserve for doubtful debts	1,320
Interest (Cr.)	340
Buildings	70,000
Motor trucks	12,000
Furniture	1,640
Debtors	15,600

	Rs.
Opening stock	15,040
Cash in hand	988
Cash at bank	14,534
Bills receivable	5,844
Purchases	85,522
Carriage inwards	1,291
Carriage outwards	800
General expenses	6,085
Insurance	783
Bad debts	613
Audit fees	400
Travelling expenses	325
Discounts	620
Sales returns	285
Investments	8,922

Prepare the trading and profit and loss account for the year ending 31st. December 2013 and a balance sheet as on that date after making the following adjustments.

(a) Stock on 31.12.2013 was valued at Rs. 15,500

(b) Depreciate motor trucks by 10% and furniture by 5%

(c) Write off a further Rs. 140 as bad debts and maintain reserve at 5% on debtors

(d) Insurance prepaid Rs. 150

(e) Interest on investment outstanding Rs. 120.

3. From the following details, prepare statement of proprietary fund with as many details as possible

Stock velocity = 6

Capital turnover ratio = 2

Fixed assets turnover ratio = 4

Gross profit turnover ratio = 20%

Debtors velocity = 2 months

Creditors velocity = 73 days.

Gross profit was Rs. 60,000. Reserves and surplus Rs. 20,000. Closing stock was Rs. 5,000 in excess of opening stock.

4. Two components A and B are consumed as follows :

Normal usage – 100 units per week each

Minimum usage – 50 units per week each

Maximum usage – 150 units per week each

Re-order quantity : A – 400 units; B – 600 units

Re-order period : A – 6 to 8 weeks B – 3 to 5 weeks.

Calculate for each component :

(a) Reorder level

(b) Minimum level

(c) Maximum level

(d) Average stock level.

5. You are given the following data for the year 2012 of the company.

Variable cost Rs. 6,00,000

Fixed cost Rs. 3,00,000

Net profit Rs. 1,00,000

Sales Rs. 10,00,000

Find

(a) P/V ratio

(b) B.E.P

(c) Profit when sales amounted to Rs. 12,00,000

(d) Sales required to earn a profit of Rs. 2,00,000.

6. With the following data for a 60% activity, prepare a budget for production at 80% and 100% activity. Production at 60% activity : 600 units.

Materials Rs. 100 per unit

Labour Rs. 40 per unit

Expenses Rs. 10 per unit

Factory expenses Rs. 40,000 (40% fixed)

Administration expenses Rs. 30,000 (60% fixed)

7. What do you mean by ABC analysis? Explain its advantages.

8. What is fund flow statement? Explain the procedure for preparing a fund flow statement.

DE–2757

Sub. Code
43

DISTANCE EDUCATION

M.C.A./MCA (Lateral) DEGREE EXAMINATION, DECEMBER 2017.

Fourth Semester

RESOURCE MANAGEMENT TECHNIQUES

(2005 to 2010 calendar year)

Time : Three hours Maximum : 100 marks

Answer any FIVE questions.

All questions carry equal marks.

($5 \times 20 = 100$)

1. (a) Write down the procedure for mathematical formulation of linear programming problem. (10)

   (b) Old hens can be bought for Rs. 300/- each but the young ones cost Rs. 500/- each. The old hens lay 3 eggs per week, each being worth Rs. 3/-. A hen costs Rs. 100 per week to feed. If I have only Rs. 8,000/- to spend for hens, how many of each kind should I buy to give a profit of more than Rs. 600/- per week, assuming that I cannot house more than 20 hens? Write a mathematical model of the above problem. (10)

2. Use penalty method to

   Minimize $Z = 4x_1 + 3x_2$

   Subject to the constraints :

   $2x_1 + x_2 \geq 10$

   $-3x_1 + 2x_2 \leq 6$

   $x_1 + x_2 \geq 6$

   $x_1 \geq 0$ and $x_2 \geq 0$

3. Four professors are each capable of teaching any one of four different courses. Class preparation time in hours for different topics varies from professor to professor and is given in the table below :

Professor	Linear programmes	Queueing theory	Dynamic programme	Regression analysis
A	2	10	9	7
B	15	4	4	8
C	13	14	16	11
D	4	15	13	9

Each professor is assigned only one course so as to minimize the total course preparation time for all course. Use Hungarian method to solve the problem.

4. (a) Discuss on M/M/1 Queueing system with infinite population model. (12)

(b) The railway marshalling yard is sufficient only for 9trains (there being 10 lines, one of which is earmarked for the shunting engine to reverse itself from the rest of the hump to the rear of the train). Trains arrive at the rate of 30 trains per day, inter-arrival time follows an exponential distribution and service time distribution is also exponential with an average of 36 minutes. Calculate the following.

(i) The probability that the yard is empty

(ii) The average time length. (8)

5. Dr STRONG is a dentist who schedules all her patients for 30 minutes appointments. Some of the patients take more or less than 30 minutes depending on the type of dental work to be done. The following summary shows the various categories of work, their probabilities and the time actually to complete the work.

Category	Time required (minutes)	Probability of category
Filling	45	0.40
Crown	60	0.15
Cleaning	15	0.15
Extraction	45	0.10
Check up	15	0.20

Simulate the dentist's clinic for four hours and determine the average waiting time for the patients as well as the idleness of the doctor. Assume that are the patients show up at the clinic at exactly their scheduled arrival time starting at 8.00 AM. Use the following random numbers handling the above problem : (20)

40 82 11 34 25 66 17 79

6. (a) Write about inventory costs and analysis. (8)

(b) The annual demand of a particular item by a company is 10,000 units. This item may be obtained from either an outside supplier or subsidiary company. The relevant data for the procurement of an item are given below :

Costs	From outside supplier (Rs.)	From subsidiary company (Rs.)
Cost per unit	12	13
Cost of placing an order	10	10
Cost of receiving an order	20	15
Storage and all carrying costs, including capital cost per unit per annum	2	2

(i) What purchase quantity and from which source would you recommend to procure?

(ii) What would be the minimum total costs in that case? (12)

7. A small project is composed of seven activities whose time estimates are listed in the table as follow :

Activity		Estimated duration (weeks)		
i	j	Optimistic	Most likely	Pessimistic
1	2	1	1	7
1	3	1	4	7
1	4	2	2	8
2	5	1	1	1
3	5	2	5	14
4	6	2	5	8
5	6	3	6	15

(a) Draw the project network.

(b) Find the expected duration and variance of each activity.

(c) Calculate early and late occurrence times for each event. What is the expected project length?

(d) Calculate the variance and standard deviation of project, Length. What is the probability that the project will be completed :

(i) At least 4 weeks earlier than expected.

(ii) No more than 4 weeks later than expected. (20)

8. A manufacturer is offered two machines A and B. A is priced at Rs. 5,000 and running costs are estimated at Rs. 800 for each of the first five years, increasing by Rs. 200 per year in the sixth and subsequent years. Machine B which has the same capacity as A, costs Rs. 2,500 but will have running costs of Rs. 1,200 per year for six years, increasing by Rs. 200 per year thereafter.

If money is worth 10% per year, which machine should be purchased? Assume that the machines will eventually be sold for scrap at a negligible price. (20)

DE–2758

Sub. Code
44

DISTANCE EDUCATION

M.C.A./M.C.A. (Lateral) DEGREE EXAMINATION, DECEMBER 2017.

Fourth Semester

OBJECT ORIENTED ANALYSIS AND DESIGN

(2005 to 2010 Calendar year)

Time : Three hours Maximum : 100 marks

Answer any FIVE questions.

All questions carry equal marks.

($5 \times 20 = 100$)

1. (a) Discuss the nature of an object with examples. (10)

   (b) How to apply the object model? Explain it. (10)

2. (a) Describe the importance of proper classification. (10)

   (b) How does object oriented approach differ from traditional approach in system development? (10)

3. (a) Write a detailed note on meta model. (10)

   (b) Discuss the overview of view process. (10)

4. (a) What is a class diagram? Explain its components and purpose in detail with example. (10)

   (b) Explain the need for object diagram. (10)

5. (a) Explain the purpose of state diagram. (10)

   (b) How do draw deployment diagram? Discuss the parameters that are controlled by these diagram.

6. (a) Describe the general approaches to classification. (10)

   (b) How is activity diagram used to model an operation? (10)

7. (a) Write a note on key abstraction mechanism. (10)

   (b) Discuss the evolution of object model. (10)

8. (a) Explain the purpose of package diagram. (10)

   (b) Describe the interplay to classes and objects. (10)

DE–2759

Sub. Code
45

DISTANCE EDUCATION

M.C.A./M.C.A. (Lateral) DEGREE EXAMINATION, DECEMBER 2017.

Fourth Semester

INTERNET PROGRAMMING

(2005 to 2010 Calendar Year)

Time : Three hours Maximum : 100 marks

Answer any FIVE questions.

All questions carry equal marks.

($5 \times 20 = 100$)

1. (a) Describe the design issue of World Wide Web. (10)

   (b) Write a note on encryption. (10)

2. (a) Explain control statements in Java Script. (10)

   (b) Discuss about the events in VB script. (10)

3. (a) List out the features of Java Script. (10)

   (b) Describe the objects in VB script. (10)

4. (a) Describe the various tags in HTML. (10)

   (b) Write a note on Lingo. (10)

5. (a) Explain Table Tag with an example. (10)

   (b) Write a note on Shokwave. (10)

6. (a) How to create an activex control to activate web page? (10)

   (b) Write a note on plugins. (10)

7. (a) What are the guidelines of graphics? Explain. (10)

   (b) Explain sound formats. (10)

8. (a) What are the key elements of web page design? Explain. (10)

   (b) Write a detailed note on animation. (10)

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DE–9836

Sub. Code
46

DISTANCE EDUCATION

M.C.A./M.C.A. (Lateral) DEGREE EXAMINATION, DECEMBER 2017.

Fourth Semester

Computer Applications

Lab VII — INTERNET PROGRAMMING

(2005 to 2010 Calendar Year)

Time : Three hours Maximum : 100 marks

Examiner should select and give ONE question to each student by LOT system.

1. (a) Write a JavaScript to create a order form to select the office articles.

   (b) Write a swing program to create buttons with

   (i) Tool tip text

   (ii) Image

   (iii) Border

   (iv) Shortcut key.

---------- Cut here ----------

2. (a) Write a VB script to do the following :

   (i) Check the given password

   (ii) Change the existing password.

   (b) Develop a web page for job recruitment agency in an IT Industry.

3. (a) Write a Java program using Applet to display any three images when three buttons in the border layout are clicked. The image should be displayed at the center.

   (b) Write a swing program to create the tabbed panels.

---------- Cut here ----------

4. (a) Write a JavaScript to create a window by using the confirm message.

   (b) Design and develop a web page for a college.

---------- Cut here ----------

5. (a) Write a VB Script to create a calendar for given month and year.

   (b) Write a Java program using Applet to create the frames and its controls.

---------- Cut here ----------

6. (a) Write a JavaScript to create a color Palet and display the text in the color chosen from the Palet with proper background color.

   (b) Create a web page in the format of front page of a newspaper using text link. Align the text with colors.

7. (a) Write a VB Script to do the following :

   (i) Display the current date.

   (ii) Find the difference between the two dates.

   (iii) Find the age of a person by providing date of birth.

   For all use the date in dd/mm/yy format.

   (b) Using atleast 20 HTML Tags, create a screen with a string "Welcome to Internet World".

---------- Cut here ----------

8. (a) Write a menu driven Java program to do database functions using database connectivity facility. The functions include display, edit, insert and delete records from a table.

   (b) Develop a single page advertisement for a shop to be opened shortly.

DE–9837

Sub. Code
47

DISTANCE EDUCATION

M.C.A./M.C.A. (Lateral) DEGREE EXAMINATION, DECEMBER 2017.

Fourth Semester

Lab VIII – NETWORK LAB

(2005 to 2010 calendar year)

Time : Three Hours Maximum : 100 Marks

Break –up of Mark

Record Note Book : 5

Algorithm/Flow chart : 15

Program : 35

Debugging and Execution : 35

Result : 10

Total : 100

Examiner should select and give ONE question to each candidate by LOT system.

1. (a) Write a Java program to add two numbers using RMI client, server techniques.

   (b) Write a Java Program to display the users and their IP addresses logged in the network.

---------- Cut here ----------

2. (a) Write a Java program to find factorial of given number using RMI client, server techniques.

   (b) Write a Java program to ENCRYPT given word.

3. (a) Write a Java program to find factorial of given number using FACTORIAL INTERFACE technique.

   (b) Write a Java Program to set up time server and time client.

---------- Cut here ----------

4. (a) Write a Java program to set up a chat server and chat client.

   (b) Write a Java program to send message from one system to another system.

---------- Cut here ----------

5. (a) Write a Java program to set up a Echo server and Echo client.

   (b) Write a Java program to read a file from the remote system.

DE–2760

Sub. Code
51

DISTANCE EDUCATION

M.C.A./M.C.A (Lateral) DEGREE EXAMINATION, DECEMBER 2017.

Fifth Semester

IMAGE PROCESSING AND ANALYSIS

(2005 to 2010 Calendar year)

Time : Three hours Maximum : 100 marks

Answer any FIVE questions.

(5 × 20 = 100)

1. (a) Explain about texture and pattern discrimination. (10)

   (b) Write a note on camera calibration. (10)

2. (a) Briefly write about sampling and quantization. (10)

   (b) Explain about Image sharpening. (10)

3. (a) Discuss about minimum mean – square error restoration. (10)

   (b) Describe about restoration by Homomorphic filtering. (10)

4. (a) Explain loss – less compression technique. (10)

   (b) Write a note on Geometric transformation. (10)

5. (a) Brief a note on stereo imaging algorithms. (10)

(b) Explain about the processing of satellite images. (10)

6. (a) Discuss about Threshold selection methods. (10)

(b) Describe the concept of pattern fitting approach. (10)

7. (a) Write about Region – based description. (10)

(b) Explain about statistical classification. (10)

8. (a) Brief a note on syntactic recognition. (10)

(b) Discuss about Graph matching. (10)

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DE–2761

Sub. Code
52

DISTANCE EDUCATION

M.C.A./M.C.A. (Lateral) DEGREE EXAMINATION, DECEMBER 2017.

Fifth Semester

SOFTWARE ENGINEERING

(2005 to 2010 Calendar Year)

Time : Three hours Maximum : 100 marks

Answer any FIVE questions only.

All questions carry equal marks.

($5 \times 20 = 100$)

1. (a) With examples, explain the different categories of software project size. (12)

   (b) Write short notes on :

   (i) Product Complexity and

   (ii) Problems of Poor Planning. (4 + 4)

2. (a) Explain the waterfall and spiral process models in detail. Also list down the merits and demerits of each. (7 + 7)

   (b) With a neat diagram, explain the hierarchical team structure. (6)

3. (a) Narrate on Structured Analysis. (6)

   (b) What is an SRS? What are the elements addressed by an SRS? Explain. (14)

4. (a) Explain the various factors which influence the software cost. (10)

   (b) Discuss the software quality attributes defined by ISO 9126. (10)

5. (a) Elaborately discuss the procedure to manage the risks involved in software development. (12)

   (b) Write a note on requirements elicitation techniques. (8)

6. (a) Write an essay on design fundamentals. (10)

   (b) How do coupling and cohesion help us in designing an effective module? (10)

7. Write notes on : $(4 \times 5 = 20)$

   (a) Unit Testing

   (b) Big Hang Approach

   (c) Debugging

   (d) Acceptance Testing.

8. (a) Explain the black box testing techniques in detail. (12)

   (b) What is cyclomatic complexity? What are the different ways to calculate the same? Explain with example. (8)

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DE–2762

Sub. Code
53

DISTANCE EDUCATION

M.C.A./M.C.A. (Lateral) DEGREE EXAMINATION, DECEMBER 2017.

Fifth Semester

VISUAL PROGRAMMING – II

(2005 – 2010 Calendar year)

Time : Three hours Maximum : 100 marks

Answer any FIVE questions.

All questions carry equal marks.

($5 \times 20 = 100$)

1. (a) Discuss the anatomy of a C++ Windows program. (10)

   (b) What is App Object? Give its importance. (10)

2. (a) Mention the views of VC ++. (10)

   (b) Explain on mainframe object. (10)

3. (a) How to connect menu items to code? (10)

   (b) Explain the technique of adding accelerator keys. (10)

4. (a) Discuss on Mouse Events. (10)

   (b) What is the need for short cut keys? How to add it? (10)

5. (a) Give the importance of message box. (10)

   (b) How to use VBX controls in code? (10)

6. (a) Explain the technique for designing dialog box. (10)

   (b) How to create a simple calculator? (10)

7. (a) Compare and contrast sequential access file with Random access file. (10)

   (b) Give the purpose of splitter windows. (10)

8. (a) What is meant by Exception Handling? How to catch multiple exceptions? (10)

   (b) How to build Active X container application? (10)

DE-2763

Sub. Code
54

DISTANCE EDUCATION

M.C.A./M.C.A. (Lateral) DEGREE EXAMINATION, DECEMBER 2017.

Fifth Semester

COMPILER DESIGN

(2005 to 2010 calendar year)

Time : Three hours Maximum : 100 marks

Answer any FIVE questions.

All questions carry equal marks.

1. (a) Explain the structure of a compiler.

   (b) Explain the algorithm for constructing an NFA equivalent to a regular expression.

2. (a) Construct a minimal state DFA for the regular expression $((a+b)(a+b))^*$.

   (b) Explain the following :

      (i) Bootstrapping a compiler

      (ii) Bookkeeping.

3. (a) Explain the following :

      (i) Parse frees

      (ii) Shift reduce parsing.

(b) (i) Write the rules for constructing FIRST and FOLLOW sets for a given grammar.

(ii) Construct FIRST and FOLLOW sets for the following grammar :

$$E \rightarrow E + T / T,\ T \rightarrow T * F / F,\ F \rightarrow (E) / id$$

4. (a) Explain the method of constructing the canonical collection of LR(0) items, with an example.

(b) Discuss using ambiguous grammars.

5. (a) Explain how port fix code can be generated using syntax directed translation scheme.

(b) Discuss the implementation of syntax directed translators.

6. (a) How scope information is represented in symbol tables? Explain.

(b) Explain the following :

(i) Lexical phase errors

(ii) Semantic errors.

7. (a) Explain the principal sources of optimization.

(b) Discuss loop optimization.

8. (a) Explain the algorithms for a simple code generator.

(b) Discuss the use of DAG in optimization.

DE–2764

Sub. Code
55

DISTANCE EDUCATION

M.C.A. (S)/M.C.A. (Lateral) DEGREE EXAMINATION, DECEMBER 2017.

Fifth Semester

DATA WAREHOUSING AND MINING

(2005 to 2010 calendar year)

Time : Three hours Maximum : 100 marks

Answer any FIVE questions.

All questions carry equal marks.

($5 \times 20 = 100$)

1. (a) Discuss about data minning vs query tools.

   (b) How data mining can be used in marketing?

2. (a) Write about OLAP Tools. Explain association rules.

   (b) What is the need for preliminary analysis of the data?

3. (a) Explain in detail about KDD.

   (b) How decision trees can be used in Data Mining?

4. (a) Describe data warehouse. Explain in detail about system process and process architecture.

   (b) Explain the concept of data marting.

5. (a) Brief about systems and data warehouse process.

   (b) Mention the role of process managers in data warehouse.

6. (a) Write about hardware architecture and physical layout of data warehouse.

   (b) How to operate the data warehouse?

7. (a) List out and explain various features of data warehouse.

   (b) How testing is done in data warehouses?

8. Write short note on :

   (a) Data mining

   (b) Knowledge discovery process

   (c) Data warehouse

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DISTANCE EDUCATION

M.C.A./M.C.A. (Lateral) DEGREE EXAMINATION, DECEMBER 2017.

Fifth Semester

Lab IX — VISUAL C++

(2005 to 2010 Calendar Year)

Time : Three hours Maximum : 100 marks

Examiner has to select and give ONE question to each candidate by LOT system.

1. (a) Write a visual C++ win32 application program using MFC that displays message "Hello Good Morning" wherever the user clicks the mouse button on the client area.

   (b) Write a visual C++ win32 application program using MFC that fills the background of the client area with a bitmap.

---------- Cut here ----------

2. (a) Write a visual C++ win32 application program using MFC that displays rounded rectangle, ellipse and polygon filled with colors.

   (b) Write a visual C++ win32 application program using MFC that displays status of CTRL, SHIFT and NUM LOCK keys.

3. (a) Write a visual C++ win32 application program using MFC that creates a list box and displays name of districts in Tamil Nadu.

   (b) Write a visual C++ win32 application program using MFC that creates two push buttons OK and CANCEL on the client area. Buttons should respond to uses click over them and display appropriate message.

---------- Cut here ----------

4. (a) Write a visual C++ win32 application program using MFC that creates a new font.

   (b) Write a visual C++ win32 application program using MFC that displays a menu. Choose the menu items using keyboard accelerator keys and display appropriate messages for the selected command in message box.

---------- Cut here ----------

5. (a) Write a visual C++ win32 application program using MFC that allows the uses to draw pictures with the help of mouse as a free hand drawing tool.

   (b) Write a visual C++ win32 application program using MFC that displays current mouse coordinates in status bar.

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DISTANCE EDUCATION

M.C.A./M.C.A. (Lateral) DEGREE EXAMINATION, DECEMBER 2017.

Fifth Semester

Computer Applications

Lab X: COMPILER DESIGN

(2005 to 2010 Calendar Year)

Time : Three hours

Maximum : 100 marks

Record Note Book : 5

Algorithm/Flow chart : 15

Program : 35

Debugging and Execution : 35

Result : 10

Total : 100

Examiner should select and give one question to each candidate by lot system.

Write program in C language for any one of the following:

1. Write a C program to construct on canonical LR parsing table for the following grammar.

$S \rightarrow CC$
$C \rightarrow cC / d$

2. Write a C program to construct predictive parsing table for the following grammar.

$S \rightarrow iCtSS_1 / a$

$S_1 \rightarrow eS / E$

$C \rightarrow b$

---------- Cut here ----------

3. Write a C program to check if a string is generated by the following grammar or not.

$E \rightarrow E + E / E * E / E - E / E / E / id / const / (E)$.

---------- Cut here ----------

4. Write a C program to recognize keywords in a source program.

---------- Cut here ----------

5. Write a C program to recognize the constants in a source program.

---------- Cut here ----------

6. Write a C program to convert an infix expression into postfix firm.

---------- Cut here ----------

7. Write a C program to translate a given expression into three address code, quadruples and triples.

---------- Cut here ----------

8. Write a C program to generate operator precedence parsing table for an operator grammar.

9. Write a C program to recognize identifiers in source program.

---------- Cut here ----------

10. Write a C program to construct left most and right most derivations for a given string using a context free grammar.

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DISTANCE EDUCATION

M.C.A./M.C.A. (Lateral) DEGREE EXAMINATION, DECEMBER 2017.

Sixth Semester

SOFTWARE PROJECT MANAGEMENT

(2005 to 2010 calendar year)

Time : Three hours Maximum : 100 marks

Answer any FIVE questions.

All questions carry equal marks.

($5 \times 20 = 100$)

1. (a) Explain about project evaluation.

   (b) Mention the problems in software projects.

2. (a) Give an overview on project analysis.

   (b) Explain software estimation.

3. (a) Write about project schedules.

   (b) How to shorten the project duration?

4. (a) What is risk management?

   (b) How to allocate the resources?

5. Write short note on :

   (a) Identifying critical activities

   (b) Planning for small projects.

6. (a) Explain about functions of software configuration management.

   (b) Discuss the standards in software configuration management.

7. (a) What is meant by prototyping?

   (b) Discuss on any one model involved in prototyping.

8. Write in detail about PRINCE project management.

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DISTANCE EDUCATION

M.C.A. (S) / M.C.A. (Lateral) DEGREE EXAMINATION, DECEMBER 2017.

Sixth Semester

MULTIMEDIA SYSTEMS

(2005 to 2010 Calendar Year)

Time : Three hours Maximum : 100 marks

Answer any FIVE questions.

All questions carry equal marks.

($5 \times 20 = 100$)

1. (a) Discuss about multimedia platforms briefly. (10)

   (b) Explain about cross platform compatibility. (10)

2. (a) Briefly explain about the different techniques used for video compression. (10)

   (b) How to evaluate a compression system? Explain. (10)

3. Write short note on the following :

   (a) Digital ink (10)

   (b) Speech audio. (10)

4. Describe about the various image compression standards in detail. (20)

5. Briefly explain about transform classes and component classes in multimedia frame works. (20)

6. Explain briefly about object oriented multimedia. (20)

7. Discuss about the various media types and media organizations briefly. (20)

8. Discuss the following

   (a) Multimedia on networks. (10)

   (b) Training and education. (10)

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DISTANCE EDUCATION

M.C.A. (S)/M.C.A. (Lateral) DEGREE EXAMINATION, DECEMBER 2017.

Sixth Semester

NEURAL NETWORKS

(2005 to 2010 Calendar Year)

Time : Three hours Maximum : 100 marks

Answer any FIVE questions.

(5 × 20 = 100)

1. (a) Write down the benefits of Artificial Neural networks. (10)

   (b) Describe about the construction of neural networks. (10)

2. (a) Discuss about the applications of neural networks. (10)

   (b) Explain about Error – correction learning. (10)

3. (a) Describe about unsupervised learning process. (10)

   (b) What is meant by adaptation? Explain. (10)

4. (a) Discuss about single layer perceptions. (10

   (b) Describe about back propagation networks. (10)

5. (a) Write a note on counter propagation networks. (10)

   (b) Discuss about associative memory. (10)

6. (a) Explain about hop field network. (10)

(b) Brief a note on bidirectional associative memory. (10)

7. (a) What is an Adaptive neural network? Explain. (10)

(b) Illustrate the architecture of ART. (10)

8. (a) Explain about optical neural network. (10)

(b) Brief a note on cognitron and neocognitron. (10)

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DISTANCE EDUCATION

M.C.A./M.C.A.(Lateral) DEGREE EXAMINATION, DECEMBER 2017.

Sixth Semester

C-SHARP (C#)

(2005 to 2010 Calendar Year)

Time : Three hours Maximum : 100 marks

Answer any FIVE questions.

(5 × 20 = 100)

1. (a) Discuss about evolution of C#. (15)

   (b) Write a note on type safety. (5)

2. (a) Explain about virtual object system.

   (b) Discuss on NGWS runtime.

3. (a) Brief about C# datatypes with example.

   (b) How to compile and run a C# application?

4. (a) Write note on Indexes and Events.

   (b) Discuss the role of constructors and destructors.

5. (a) Explain about Iteration statements with example.

   (b) How to write the components in C#?

6. (a) Discuss the components of .Net.

   (b) What is meant by Time Safety?

7. (a) How to create a client?

   (b) Explain the technique of working with Namespaces.

8. (a) Discuss the services provided for platform invocation.

   (b) Explain the concept of Intermediate Language Disassembler.

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DISTANCE EDUCATION

M.C.A./M.C.A. (Lateral) DEGREE EXAMINATION, DECEMBER 2017.

Sixth Semester

Computer Applications

LAB XI — MULTIMEDIA APPLICATIONS

(2005 to 2010 Calendar year)

Time : Three hours Maximum : 100 marks

Examiner should select and give ONE question to each candidate by lot system.

1. (a) Write a C++ program to compress the image using contour coding technique.

Given image :

$$\begin{bmatrix} 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 2 & 2 & 2 & 2 & 2 & 2 & 2 \\ 1 & 1 & 1 & 1 & 1 & 1 & 1 & 2 & 2 & 2 & 2 & 2 & 2 & 2 & 2 \\ 1 & 1 & 1 & 1 & 1 & 1 & 2 & 2 & 2 & 2 & 2 & 2 & 2 & 2 & 2 \\ 1 & 1 & 1 & 1 & 1 & 1 & 1 & 2 & 2 & 2 & 2 & 2 & 2 & 2 & 2 \\ 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 2 & 2 & 2 & 2 \end{bmatrix}$$

(b) Design a birthday greeting page including various graphic features in falsh.

2. (a) Write a C++ program for convolution using given template.

Image

$$\begin{pmatrix} 1 & 1 & 3 & 3 & 4 \\ 1 & 1 & 4 & 4 & 3 \\ 2 & 1 & 3 & 3 & 3 \\ 1 & 1 & 1 & 4 & 4 \end{pmatrix}$$

Template

$$\begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$$

(b) Design an animated cartoon picture.

---------- Cut here ----------

3. (a) Write a C++ program for creating an image from the given image using automatic selection of gray level for splitting technique.

$$\begin{pmatrix} 1 & 8 & 4 & 3 & 6 & 2 & 5 & 2 & 8 & 4 & 6 & 2 & 5 \\ 0 & 3 & 8 & 3 & 6 & 5 & 4 & 0 & 3 & 8 & 3 & 8 & 7 \\ 0 & 9 & 8 & 0 & 5 & 4 & 8 & 5 & 9 & 3 & 7 & 2 & 9 \end{pmatrix}$$

(b) Do the image animation using Flash.

---------- Cut here ----------

4. (a) Write a C++ program to create a second image array model from the given image using gray-level transformation bunching technique.

$$\begin{pmatrix} 1 & 8 & 4 & 3 & 6 & 2 & 5 & 2 & 8 & 4 & 6 & 2 & 5 \\ 0 & 3 & 8 & 3 & 6 & 5 & 4 & 0 & 3 & 8 & 3 & 8 & 7 \\ 0 & 9 & 8 & 0 & 5 & 4 & 8 & 5 & 9 & 3 & 7 & 2 & 9 \end{pmatrix}$$

Also draw histograms for the first and second images.

(b) Design a sales advertisement for a product using Flash.

5. (a) Write a C++ program for two dimensional image animation using computer graphics.

   (b) Prepare a multimedia presentation showing the features of the courses offered in an university.

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6. (a) Write a C++ program for grey level transformation techniques.

   (i) Thresholding

   (ii) Bunching.

   (b) Prepare a multimedia presentation shows the features of a software package developed by you.